

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2015

Subject: 394171 Mathematics I

Section: 15-16

Date: 13 October 2015

Time 8:00-10:00

Name \_\_\_\_\_ ID \_\_\_\_\_ Class \_\_\_\_\_

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Student who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks and dictionaries are **NOT** allowed.
4. Calculators are **NOT** allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. This examination has 6 pages (including this page), 9 questions and a total score of 90 marks.
7. Write all your answers in this examination sheet.

**Question 1.** Given  $U = \{2, 3, 4, \dots, 10\}$ ,  $A = \{2, 4, 6\}$ ,  $B = \{3, 4, 5, 6, 7\}$  and  $C = \{3, 5, 7, 9\}$ . Find the set  $(A - C)' \cap B$  and the number of element of  $P((B - C) \cup A)$ . (10 marks)

**Question 2.** . Given  $A$ ,  $B$ , and  $C$  are any sets such that  $n(B) = 42$ ,  $n(C) = 28$ ,  $n(A \cap C) = 8$ ,  $n(A \cap B \cap C) = 3$ ,  $n(A \cap B \cap C') = 2$ ,  $n(A \cap B' \cap C') = 20$  and  $n(A \cup B \cup C) = 80$ . How many elements of the set  $A' \cap B \cap C$ . (10 marks)

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**Question 3.** Given  $A$ ,  $B$ , and  $C$  are any sets such that  $n(A \cup B) = 16$ ,  $n(A) = 8$ ,  $n(C) = 5$ , and  $n(A \cap B \cap C) = 2$ . Evaluate all possible number of elements in the set  $(A \cap B) \times C, (n((A \cap B) \times C))$ . (10 marks)

**Question 4.** Find the set of all solutions of the inequality  $x^3 + 6 < 2x^2 + 5x$   
(10 marks)

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Question 5. Solving the inequality  $\frac{1}{x+1} + \frac{2}{x+3} \geq \frac{3}{x+2}$ . (10 marks)

Question 6 6.1 Find an identity of the operator,  $x \oplus y = x + y + 5$ , where  $x, y$  are real numbers. (5 marks)

6.2 Find the inverse of  $\frac{1}{5}$  of the above operator  $x \oplus y$ . (5 marks)

**Question 7** Given the relation  $r = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y \geq x^2 + 2x, y \leq 4 - x\}$ . Sketch the graph of the relation and find its domain and range. (10 marks)

**Question 8** Given  $f\left(\frac{2x+1}{2}\right) = 4x - 5$ , Find the expression  $f^{-1}(2x+1) - f^{-1}(2x-1)$ .  
(10 marks)

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Question 9 Given  $f(2x-1) = 4x+3$ ,  $g(x) = \frac{x}{x+2}$  and  $h(x-1) = \sqrt{x}$ . Find  $(f^{-1} \circ g)(x)$  and  $(h^{-1} \circ g)(x)$ . (10 marks)

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